

Exponential Relaxation of Fisher Path Speed on the Statistical Manifold of Learned Transition Matrices: Numerical Evidence from Quantum Processors

Continuation of Grammar Fingerprinting (doi:10.5281/zenodo.19158088), Fisher Information Threshold Study (doi:10.5281/zenodo.19391582), LLM Grammar Fingerprinting (doi:10.5281/zenodo.19454201), Curvature at the Fisher Threshold (Research Note, April 2026)

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Abstract

We study the dynamics of learned transition matrices on the Fisher–Rao statistical manifold as a function of data length N , using the Grammar Fingerprinting pipeline applied to 28 Google Sycamore quantum processor readout configurations. By computing the Fisher path speed—the rate of change of the transition matrix $T(N)$ on the manifold, measured via the Fisher information metric—we show that the speed decays exponentially: $v(N) = v_{\infty} + A \cdot \exp(-N/\tau)$. This exponential relaxation is the dominant model in 25–26/28 readout configurations across all tested alphabet sizes $K \in \{5, 7, 9\}$, and the relaxation timescale τ is systematically larger than the previously reported Fisher trace threshold N^* by a factor of $\tau/N^* \approx 3.2$ (median, $K=7$). The product $A \times \tau$ is stable across all three Sycamore topologies (1D Snake, 2D Block, Bulk Full) within each K , with overall coefficients of variation 0.14–0.17 across $K = 5, 7, 9$. Varying K , $A \times \tau$ scales as a power law: $A \times \tau \approx 21.5 \cdot K^{0.57}$, with residuals below 1.4% for all three K values. We compare the functional form of the Fisher speed profile with the macroscopicity threshold derived by Coppo, Pranzini and Verrucchi (2022) for the emergence of classical-like behavior in the Page–Wootters framework: both exhibit exponential decay with a logarithmic threshold dependence on the tolerance parameter. The Grammar Fingerprinting threshold N^* and the Coppo threshold N_t are identified as two distinct points on the same exponential curve—the onset of detectable structure (~27% decay) and the completion of classical distinguishability (~95% decay), respectively. The formal derivation of this correspondence remains an open problem.

1. Introduction

The Grammar Fingerprinting methodology (Csaplár, 2026a) established that quantum processor readout sequences carry statistically learnable temporal structure, detectable via the Fisher information trace of a SAX-discretized, LSTM-learned transition matrix. The Fisher Information Threshold Study (Csaplár, 2026b) showed that a data-length threshold N^* exists below which no reliable structure is detectable, and that this threshold is consistent across 28 Sycamore readout configurations, 3 LSTM seeds, and broadly aligned with the ~8,000-point classification threshold from supervised experiments. The LLM extension (Csaplár, 2026c) demonstrated domain agnosticism: the same pipeline detects temporal grammar in large language model entropy series with signal-to-noise ratios of 2.2–263×. The Curvature Research Note (Csaplár, 2026d) introduced a second characteristic scale N_{curv} (the maximum of $|d^2\text{Tr}(F)/dN^2|$) and conjectured a geometric phase transition on the statistical manifold at the threshold.

All prior work analyzed the Fisher trace $\text{Tr}(F)$ —a scalar summary of the learned transition matrix $T(N)$. The present study moves beyond scalars to the geometry of the full transition matrix manifold. We compute the saved $T(N)$ matrices for all (readout, N) pairs, embed them in the Fisher–Rao manifold of row-stochastic matrices, and measure the Fisher path speed: the norm of the manifold velocity dT/dN ,

evaluated in the Fisher information metric. This provides a direct, geometry-aware measure of how rapidly the learned grammar changes as data accumulates.

We find that the Fisher path speed decays exponentially in N , with a characteristic timescale τ that is robust across topologies. The product $A \times \tau$ (amplitude times timescale) is an empirical invariant within a given K , and scales as a power law across K . We compare this relaxation profile with the macroscopicity threshold of Coppo, Pranzini and Verrucchi (2022), who derived the minimum system size N_t for the emergence of classical-like behavior in the Page–Wootters mechanism via generalized coherent state discrimination.

2. Methods

2.1 Transition matrix extraction and storage.

The Grammar Fingerprinting pipeline (SAX alphabet K , LSTM hidden_dim=32, seq_len=16, epochs=50) is applied to each of the 28 Sycamore readout configurations on a data-length grid of 15 points ($N = 500, 1000, 2000, \dots, 40,000$). For each (readout, N) pair, the $K \times K$ row-stochastic transition matrix $T(N)$ is saved, yielding 420 matrices per K value. Three alphabet sizes are tested: $K \in \{5, 7, 9\}$.

2.2 Fisher–Rao embedding.

Each row of $T(N)$ defines a point on the $(K-1)$ -simplex. The Fisher information metric on the multinomial simplex in free coordinates $\theta = (p_1, \dots, p_{K-1})$ is $G_{ab} = \delta_{ab}/p_a + 1/p_K$. The full $K \times K$ transition matrix lives on a product of K independent $(K-1)$ -simplices, giving a Riemannian product manifold with block-diagonal Fisher metric. The total number of free parameters—and thus the dimension of the embedding manifold—is $d = K \times (K-1)$, i.e. $d = 20, 42, 72$ for $K = 5, 7, 9$ respectively. This is the intrinsic dimension of the product-simplex manifold, not an ambient embedding dimension. Two manifold quantities are computed for each (readout, N) pair: $\log \det G(N)$ (the log-determinant of the Fisher metric tensor at $T(N)$), and fisher_speed_dn (the Fisher path speed $\|d\theta/dN\|_G$, estimated via finite differences between consecutive $T(N)$ matrices).

2.3 Exponential relaxation fit.

For each readout, the Fisher speed profile $v(N)$ is fitted to two models: exponential $v = v_\infty + A \cdot \exp(-N/\tau)$, and power law $v = v_\infty + a \cdot N^{-b}$. Model selection is by AIC. The product $A \times \tau$ is recorded for each readout. The grammar threshold N^* is taken from the Fisher trace maximum on the same N -grid.

3. Results

3.1 Exponential dominance.

The exponential model is preferred (lower AIC) in 25–26 of 28 readouts for all three K values: 26/28 at $K=5$, 25/28 at $K=7$, 25/28 at $K=9$. The Fisher path speed universally shows high values at small N (rapid exploration of the manifold) followed by monotonic decay toward a near-zero asymptote.

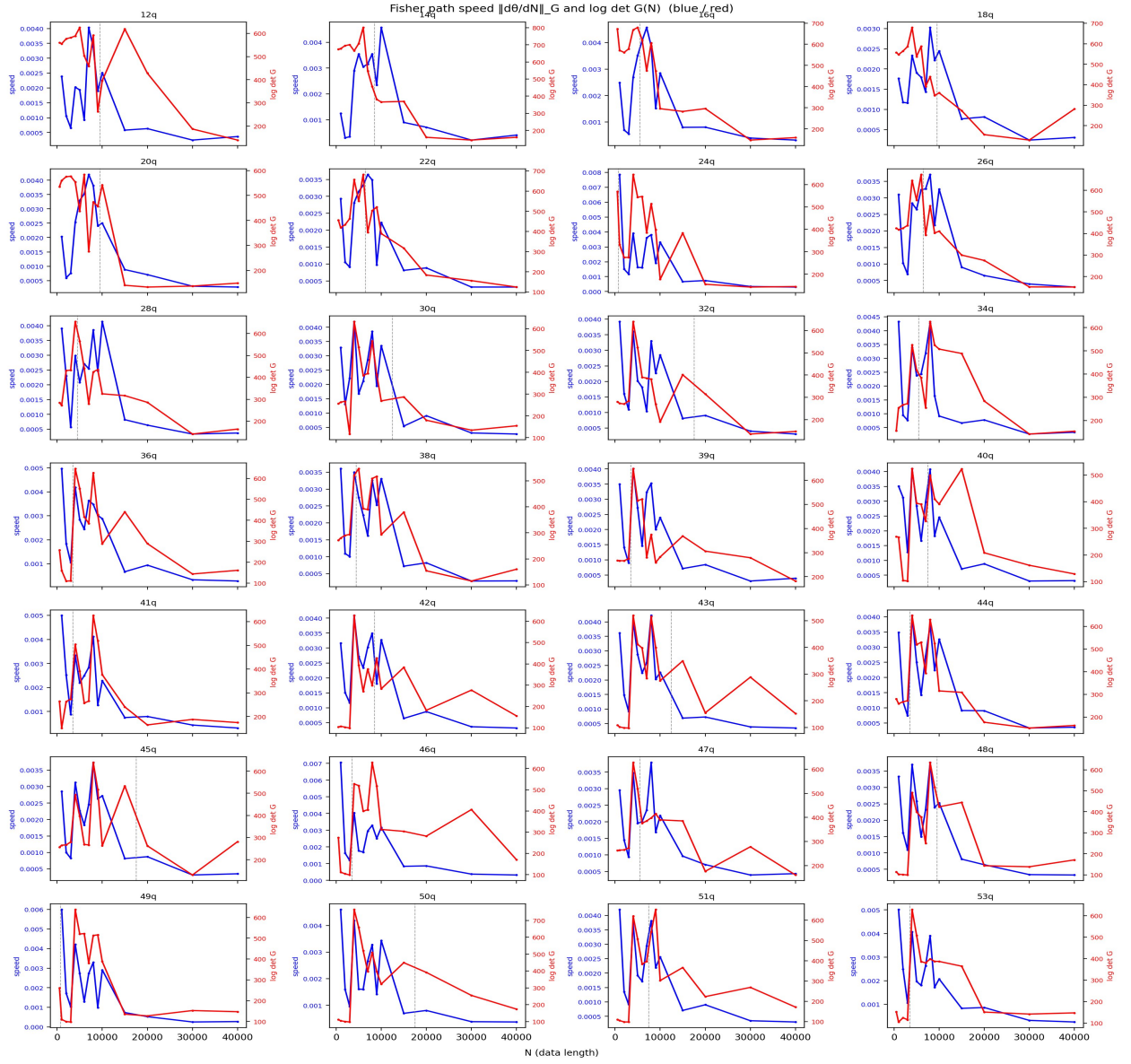


Figure 1: Fisher path speed (blue, left axis) and $\log \det G$ (red, right axis) versus data length N for all 28 Sycamore readout configurations ($K=7$). Dashed vertical lines indicate the estimated grammar threshold N^* . Speed decays from high values at small N toward near-zero at large N , consistent with exponential relaxation.

3.2 Relaxation timescale and threshold.

At $K=7$, the median relaxation timescale is $\tau \approx 22,400$ (range 13,800–39,700). The ratio τ/N^* has a median of 3.22, meaning N^* sits at $N^*/\tau \approx 0.31$ on the exponential curve—where only $\sim 27\%$ of the total decay has occurred. N^* thus marks the onset of detectable structure, not the completion of relaxation.

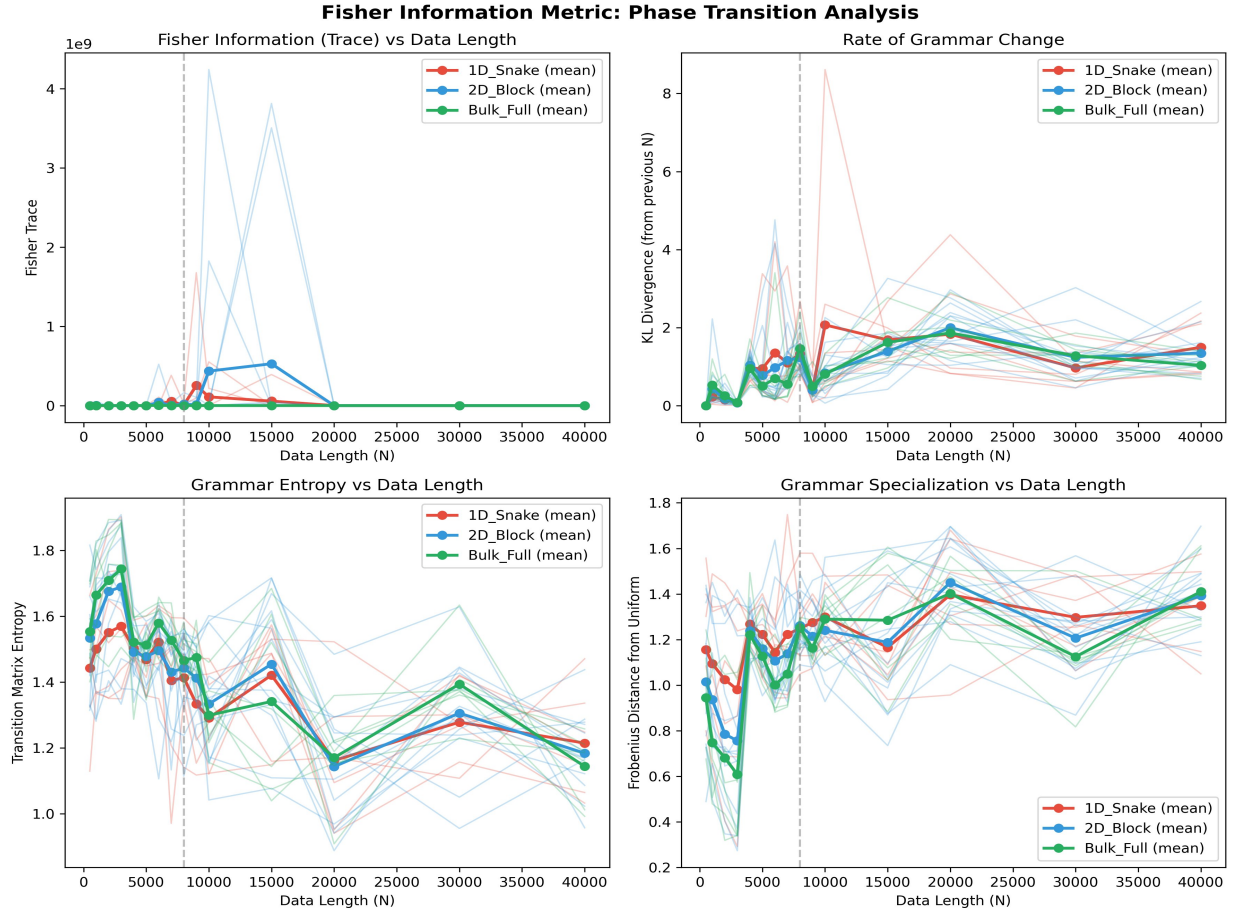


Figure 2: Fisher Information phase transition analysis for 28 Sycamore readouts ($K=7$). Upper left: Fisher trace vs N , showing topology-dependent peaks around the $\sim 8,000$ threshold. Upper right: KL divergence between consecutive $T(N)$ matrices. Lower left: transition matrix entropy, decreasing as grammar specializes. Lower right: Frobenius distance from uniform, increasing with specialization. Dashed vertical line marks the $\sim 8,000$ -point reference threshold.

3.3 $A \times \tau$ stability within K .

Table 1: $A \times \tau$ invariance across topologies ($K=7$, Sycamore).

Topology	n	Median $A \times \tau$	Std	CV
1D Snake	6	66.04	15.32	0.23
2D Block	13	65.87	9.70	0.15
Bulk Full	6	64.95	8.47	0.13
All	25	65.87	10.73	0.16

Table 1 shows $A \times \tau$ for $K=7$ (detailed breakdown). The median is 65.87 across all 25 exponential-fit readouts. The three topologies agree within 2%. The same topology-independence holds at $K=5$ (overall CV = 0.17) and $K=9$ (overall CV = 0.14), confirming that $A \times \tau$ stability is not specific to $K=7$. This suggests that $A \times \tau$ is a property of the manifold geometry and the learning pipeline, not of the physical hardware topology.

3.4 Power law scaling across K .

Table 2: $A \times \tau$ scaling across K (28 Sycamore readouts, exponential fits).

K	$d = K(K-1)$	n_{exp}	Median $A \times \tau$	Predicted*	Error
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5	20	26	53.32	53.64	0.6%
7	42	25	65.87	64.94	1.4%
9	72	25	74.30	74.90	0.8%

*Power law fit: $A \times \tau = 21.5 \cdot K^{0.57}$; all residuals < 1.4%. n_exp: number of readouts where exponential model is preferred by AIC.

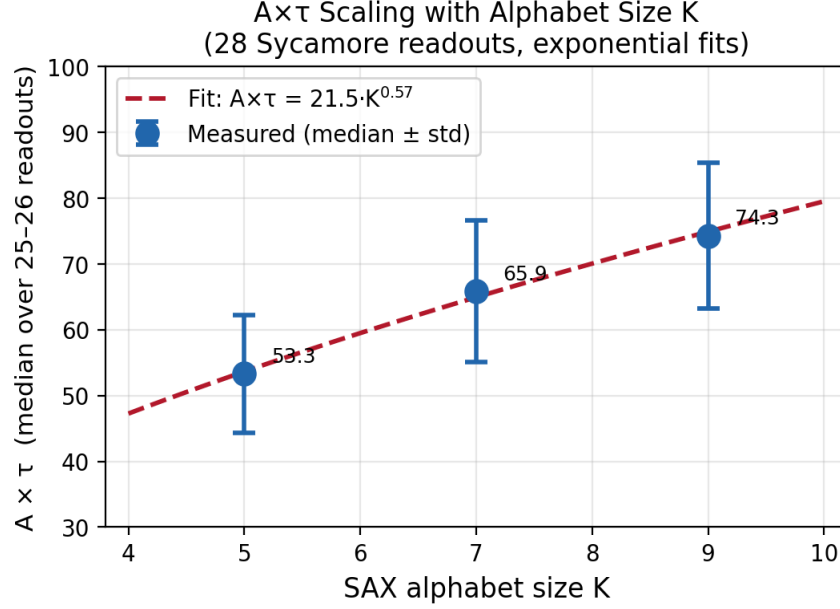


Figure 3: $A \times \tau$ (median $\pm$ std across readouts) versus SAX alphabet size K for $K = 5, 7, 9$. Dashed red line: power law fit $A \times \tau = 21.5 \cdot K^{0.57}$. The growth is sublinear in both K and the manifold dimension $d = K(K-1)$.

3.5 LLM entropy series (exploratory).

The same analysis applied to LLM entropy time series (Qwen2.5-Coder:7B, $K=7$, 4 regimes $\times$ 3 seeds = 12 conditions) yields noisier results. Two conditions (mathematical, seeds 7 and 99) are excluded because the exponential fit converges to the parameter boundary ($\tau \rightarrow 500,000$, effectively infinite), indicating numerical instability rather than meaningful relaxation. Of the remaining 10 conditions, the power-law model is preferred in all cases, consistent with the multi-scale temporal structure documented in the Curvature Research Note ($N_{\text{curv}} \neq N^*$ separation in multi-scale regimes). The median $A \times \tau$ from secondary exponential fits is 91.5 (one-sample t-test vs. 65.97: $p = 0.11$), which does not reject consistency with the Sycamore $K=7$ value but is not confirmatory. LLM results are reported as exploratory and are not used in the main analysis.

4. Comparison with the Coppo–Pranzini–Verrucchi Threshold

4.1 The macroscopicity threshold.

Coppo, Pranzini and Verrucchi (2022) derived a threshold system size $N_t(\epsilon, \delta)$ beyond which a quantum system admits a classical-like description via generalized coherent state (GCS) discrimination. For the $\text{su}(2)$ case: $N_t = \ln(\epsilon) / \ln[\cos(\delta/2)]$, where ϵ is a tolerance parameter and δ is the experimental resolution. The GCS overlap $|\langle \Omega | \Omega' \rangle|^2 = [\cos(\theta/2)]^{2N}$ decays exponentially in N .

4.2 Functional form comparison.

The overlap speed— $d/dN |\langle \Omega | \Omega' \rangle|^2$ —is a pure exponential: $v_{\text{Coppo}}(N) = A_C \cdot \exp(-N/\tau_C)$, with $\tau_C = -1/(2 \cdot \ln[\cos(\delta/2)])$. This has the same functional form as the Grammar Fingerprinting Fisher speed: $v(N) = v_\infty + A \cdot \exp(-N/\tau)$. Both are exponential relaxations with a single characteristic timescale. The

threshold dependence on tolerance is logarithmic in both cases: $N^*(\epsilon) = -\tau \cdot \ln(\epsilon/A)$ for Grammar Fingerprinting, $N_t(\epsilon) = \ln(\epsilon)/\ln[\cos(\delta/2)]$ for Coppo et al.

4.3 Two thresholds on one curve.

The Grammar Fingerprinting threshold N^* sits at $N^*/\tau \approx 0.31$, where the speed has decayed to $\sim 73\%$ of its initial value—the onset of detectable structure. The Coppo threshold N_t (at $\epsilon = 0.22$, their worked example) sits at $N_t/\tau \approx 3.03$, where the overlap has decayed to $\sim 5\%$ —the completion of classical distinguishability. Both thresholds live on the same exponential decay curve but mark different physical milestones: structure onset vs. classical completion. Their ratio is $N_t/N^* \approx 9.7$ for $\epsilon = 0.22$.

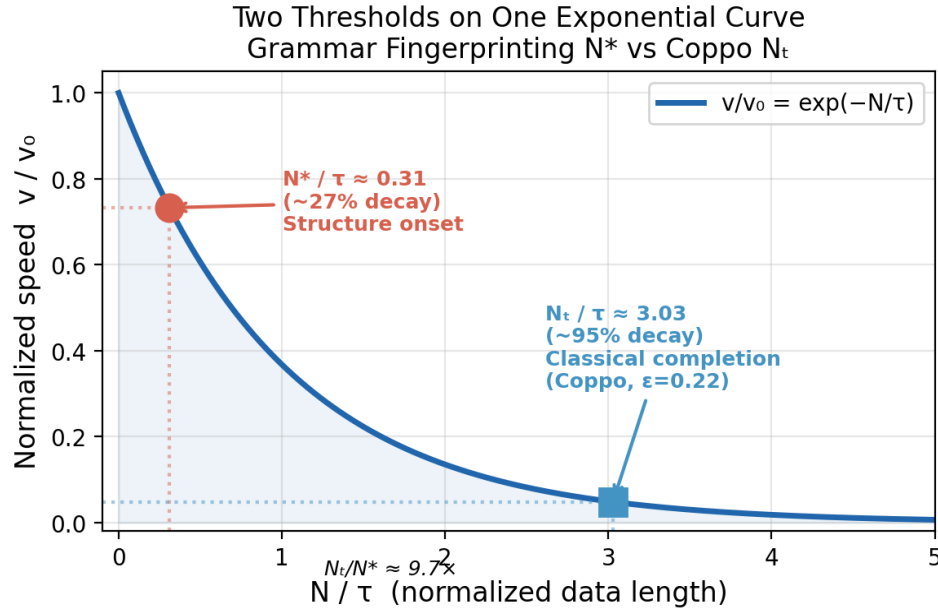


Figure 4: Normalized exponential decay curve $v/v_0 = \exp(-N/\tau)$. The Grammar Fingerprinting threshold N^* (red circle) sits at $N^*/\tau \approx 0.31$ ($\sim 27\%$ decay, structure onset). The Coppo threshold N_t (blue square) sits at $N_t/\tau \approx 3.03$ ($\sim 95\%$ decay, classical completion). Both are points on the same curve; their ratio is $\sim 9.7\times$.

4.4 What matches, what does not.

The functional form (exponential relaxation $\rightarrow$ logarithmic threshold) matches. The topology independence of $A \times \tau$ within a given K mirrors the algebra-independence of the Coppo ϵ -dependence (always proportional to $\ln \epsilon$). However, the Coppo invariant $A_C \times \tau_C = 1/2$ (on S_2) does not hold for Grammar Fingerprinting ($A \times \tau \approx 66$ at $K=7$, scaling as $K^{0.57}$). This is expected: the Coppo result applies to the $SU(2)$ coherent state manifold S_2 (intrinsic dimension 2), while Grammar Fingerprinting operates on a product of K independent $(K-1)$ -simplices (intrinsic dimension $d = K(K-1)$). The correspondence is structural (shared functional form and threshold logic), not parametric (different manifold, different invariant).

5. Discussion

The main finding is that the Grammar Fingerprinting pipeline, viewed geometrically, exhibits a universal relaxation dynamic: the learned transition matrix approaches a fixed point on the Fisher–Rao manifold exponentially in N , with a timescale τ that is topology-independent and $\sim 3\times$ larger than the Fisher trace threshold N^* . The product $A \times \tau$ is an empirical invariant of the pipeline at fixed K , invariant to hardware topology (CV = 0.14–0.17 across $K = 5, 7, 9$), and scales sublinearly ($\sim K^{0.57}$) with alphabet size.

The Coppo comparison provides a theoretical frame: both systems exhibit exponential distinguishability decay, and both define thresholds as logarithmic functions of a tolerance. The Grammar Fingerprinting threshold N^* corresponds to structure onset (early on the decay curve), while the Coppo threshold N_t corresponds to classical completion (late on the curve). Whether this analogy reflects a deeper formal connection—perhaps via the information geometry of the transition-matrix manifold and the Page–Wootters mechanism—remains an open problem, consistent with the conjecture stated in the Curvature Research Note.

The sublinear K -scaling ($A \times \tau \sim K^{0.57}$) requires further investigation. Three K -values suffice to identify the scaling as sublinear but not to determine its functional form unambiguously. Additional K -values (e.g., $K = 4, 6, 8, 11$) would clarify whether the exponent is robust or an artifact of the limited sample. An analytical derivation of $A \times \tau$ from the Fisher–Rao geometry of the simplex product would be a strong test of the geometric interpretation.

6. Summary

1. The Fisher path speed of Grammar Fingerprinting transition matrices decays exponentially in data length N , with the exponential model preferred in 25–26/28 readouts across $K = 5, 7, 9$.
2. The relaxation timescale τ is systematically $\sim 3.2\times$ larger than the Fisher trace threshold N^* , placing N^* at the onset ($\sim 27\%$ decay) of the exponential relaxation.
3. The product $A \times \tau$ is topology-independent within each K (CV 0.14–0.17 for all three tested K values) and scales as $A \times \tau \approx 21.5 \cdot K^{0.57}$ across $K = 5, 7, 9$ (residuals $< 1.4\%$).
4. The exponential relaxation functional form is shared with the Coppo–Pranzini–Verrucchi macroscopicity threshold. N^* and N_t are two points on the same exponential curve, marking structure onset and classical completion respectively.
5. The formal connection between the Fisher–Rao geometry of transition matrices and the Page–Wootters mechanism remains an open problem.

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Data and Code Availability

Sycamore readout data: Dryad DOI 10.5061/dryad.k6t1rj8. Grammar Fingerprinting code and all results: https://github.com/csaplard/QuantumCircuit_Grammar_Research. Transition matrices ($K=5, 7, 9$), Fisher embedding CSVs, speed fit summaries, and LLM entropy results are maintained in the project repository.

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